

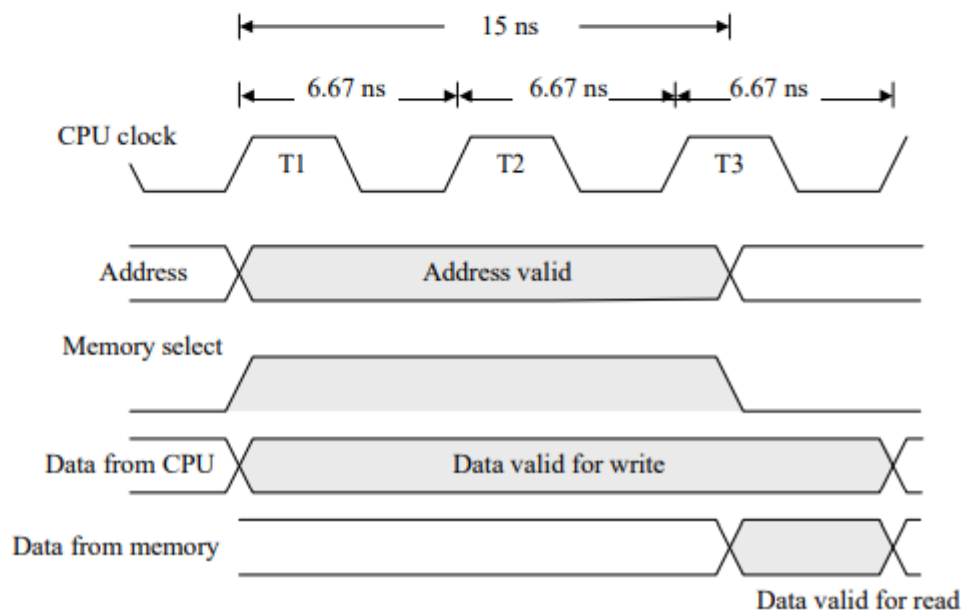
- 7.1**
- (a) $32\text{ K} \times 64$
 $32\text{ K} = 2^5 \times 2^{10} = 2^{15}$. $A = 15$ $D = 64$
- (b) $4\text{ G} \times 8$
 $4\text{ G} = 2^2 \times 2^{30} = 2^{32}$. $A = 32$ $D = 8$
- (c) $8\text{ M} \times 32$
 $8\text{ M} = 2^3 \times 2^{20} = 2^{23}$. $A = 23$ $D = 32$
- (d) $128\text{ K} \times 16$
 $128\text{ K} = 2^7 \times 2^{10} = 2^{17}$. $A = 17$ $D = 16$

- 7.2** (a) 2^{18} (b) 2^{32} (c) 2^{25} (d) 2^{18}

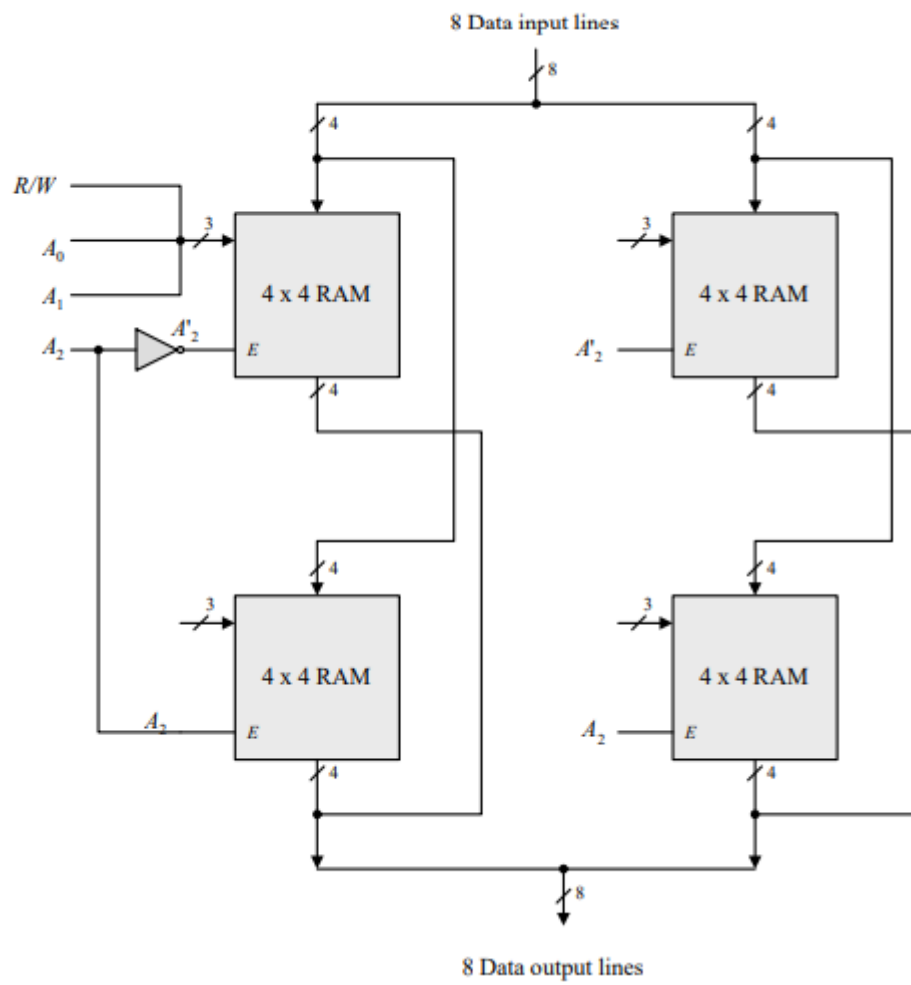
- 7.3** Address: $565_{10} = 10_0011_0101_2$

Data word: $1,210_{10} = 0000_0100_1011_1010_2$

- 7.4** $f_{\text{CPU}} = 150\text{ MHz}$, $T_{\text{CPU}} = 1/f_{\text{CPU}} = 6.67^{-9}\text{ Hz}^{-1}$



7.6



- 7.7 (a) $4\text{ K} = 2^{12} = 2^6 \times 2^6 = 64 \times 64$
 Size of each decoder is 6×64
 Number of AND gates = $64 + 64 = 128$

(b) $2017_{10} = 11111100001_2$
 $x = 31 \quad y = 33$

- 7.8 (a) Memory capacity of a $128\text{ K} \times 8\text{ RAM chip} = 128\text{ K} \times 1\text{ byte} = 128\text{ K bytes}$
 Number of chips = $2\text{ M} / 128\text{ K} = 16$

(b) $2\text{ M} = 2^{21}$. So 21 address lines will be required.
 $128\text{ K} = 2^{17}$. So 17 lines are connected to address input of each chip.

(c) Number of lines for the chip select inputs = $21 - 17 = 4$. Size of decoder is 4×16 .

7.9 Number of address lines = $10 + 12 = 24$. Memory capacity = 2^{24} words.

7.10 Data Word = 1101 0011

Bit Position	1	2	3	4	5	6	7	8	9	10	11	12	13
	P_1	P_2	1	P_4	1	0	1	P_8	0	0	1	1	P_{13}
Composite word	0	1	1	1	1	0	1	0	0	0	1	1	1

$$P_1 = \text{XOR}(3, 5, 7, 9, 11) = 1 \oplus 1 \oplus 1 \oplus 0 \oplus 1 = 0$$

$$P_2 = \text{XOR}(3, 6, 7, 10, 11) = 1 \oplus 0 \oplus 1 \oplus 0 \oplus 1 = 1$$

$$P_4 = \text{XOR}(5, 6, 7, 12) = 1 \oplus 0 \oplus 1 \oplus 1 = 1$$

$$P_8 = \text{XOR}(9, 10, 11, 12) = 0 \oplus 0 \oplus 1 \oplus 1 = 0$$

$$P_{13} = \text{XOR}(1, \dots, 12) = 1$$

$$\text{Composite 13-word code} = 0\ 1111\ 0100\ 0111$$

7.11 Data word = 11001001010.

Bit Position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	P_1	P_2	1	P_4	1	0	0	P_8	1	0	0	1	0	1	0
Composite word	1	0	1	1	1	0	0	1	1	0	0	1	0	1	0

$$P_1 = \text{XOR}(3, 5, 7, 9, 11, 13, 15) = 1 \oplus 1 \oplus 0 \oplus 1 \oplus 0 \oplus 0 \oplus 0 = 1$$

$$P_2 = \text{XOR}(3, 6, 7, 10, 11, 14, 15) = 1 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 1 \oplus 0 = 0$$

$$P_4 = \text{XOR}(5, 6, 7, 12, 13, 14, 15) = 1 \oplus 0 \oplus 0 \oplus 1 \oplus 0 \oplus 1 \oplus 0 = 1$$

$$P_8 = \text{XOR}(9, 10, 11, 12, 13, 14, 15) = 1 \oplus 0 \oplus 0 \oplus 1 \oplus 0 \oplus 1 \oplus 0 = 1$$

$$\text{Composite 15-bit code word} = 101\ 1100\ 1100\ 1010$$

7.12 (a) 0000 1010 1010

Bit Position	1	2	3	4	5	6	7	8	9	10	11	12
Composite word	0	0	0	0	1	0	1	0	1	0	1	0
	P_1	P_2	0	P_4	1	0	1	P_8	1	0	1	0

$$C_1 = \text{XOR}(1,3,5,7,9,11) = 0 \oplus 0 \oplus 1 \oplus 1 \oplus 1 \oplus 1 = 0$$

$$C_2 = \text{XOR}(2,3,6,7,10,11) = 0 \oplus 0 \oplus 0 \oplus 1 \oplus 0 \oplus 1 = 0$$

$$C_4 = \text{XOR}(4,5,6,7,12) = 0 \oplus 1 \oplus 0 \oplus 1 \oplus 0 = 0$$

$$C_8 = \text{XOR}(8,9,10,11,12) = 0 \oplus 1 \oplus 0 \oplus 1 \oplus 0 = 0$$

$C = 0000$. No error.

Correct data after removing P_1, P_2, P_4 and P_8 is 0101 1010.

(b) 0111 0110 1111

Bit Position	1	2	3	4	5	6	7	8	9	10	11	12
Composite word	0	1	1	1	0	1	1	0	1	1	1	1
	P_1	P_2	1	P_4	0	1	1	P_8	1	1	1	1

$$C_1 = \text{XOR}(1,3,5,7,9,11) = 0 \oplus 1 \oplus 0 \oplus 1 \oplus 1 \oplus 1 = 0$$

$$C_2 = \text{XOR}(2,3,6,7,10,11) = 1 \oplus 1 \oplus 1 \oplus 1 \oplus 1 \oplus 1 = 0$$

$$C_4 = \text{XOR}(4,5,6,7,12) = 1 \oplus 0 \oplus 1 \oplus 1 \oplus 1 = 0$$

$$C_8 = \text{XOR}(8,9,10,11,12) = 0 \oplus 1 \oplus 1 \oplus 1 \oplus 1 = 0$$

$C = 0000$. No error.

Correct data after removing P_1, P_2, P_4 and P_8 is 1011 1111.

(c) 1000 1010 0011

Bit Position	1	2	3	4	5	6	7	8	9	10	11	12
Composite word	1	0	0	0	1	0	1	0	0	0	1	1
	P_1	P_2	0	P_4	1	0	1	P_8	0	0	1	1

$$C_1 = \text{XOR}(1,3,5,7,9,11) = 1 \oplus 0 \oplus 1 \oplus 1 \oplus 0 \oplus 1 = 0$$

$$C_2 = \text{XOR}(2,3,6,7,10,11) = 0 \oplus 0 \oplus 0 \oplus 1 \oplus 0 \oplus 1 = 0$$

$$C_4 = \text{XOR}(4,5,6,7,12) = 0 \oplus 1 \oplus 0 \oplus 1 \oplus 1 = 1$$

$$C_8 = \text{XOR}(8,9,10,11,12) = 0 \oplus 0 \oplus 0 \oplus 1 \oplus 1 = 0$$

$C = 0100$. Error in bit 4 = P_4 .

Correct data after removing P_1, P_2, P_4 and P_8 is 0101 0011.

7.13

(a) 8 bits

The relation: $n + k \leq 2k - 1$, is satisfied for $k = 4$, since $8 + 4 \leq 2^4 - 1$

1 extra bit for single-error correction and double-error detection.

So, 5 check bits are required.

(b) 24 bits

The relation: $n + k \leq 2k - 1$, is satisfied for $k = 5$, since $24 + 5 \leq 2^5 - 1$

1 extra bit for single-error correction and double-error detection.

So, 6 check bits are required.

(c) 56 bits

The relation: $n + k \leq 2k - 1$, is satisfied for $k = 6$, since $56 + 6 \leq 2^6 - 1$

1 extra bit for single-error correction and double-error detection.

So, 7 check bits are required.

7.14 (a)

Bit Position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	P_1	P_2	1	P_4	0	0	1	P_8	0	0	1	1	0	1	P_{15}
Composite word	1	0	1	1	0	0	1	1	0	0	1	1	0	1	P_{15}

$$P_1 = \text{XOR}(3,5,7,9,11,13) = 1 \oplus 0 \oplus 1 \oplus 0 \oplus 1 \oplus 0 = 1$$

$$P_2 = \text{XOR}(3,6,7,10,11,14) = 1 \oplus 0 \oplus 1 \oplus 0 \oplus 1 \oplus 1 = 0$$

$$P_4 = \text{XOR}(5,6,7,12,13,14) = 0 \oplus 0 \oplus 1 \oplus 1 \oplus 0 \oplus 1 = 1$$

$$P_8 = \text{XOR}(9,10,11,12,13,14) = 0 \oplus 0 \oplus 1 \oplus 1 \oplus 0 \oplus 1 = 1$$

$$14\text{-bit Composite code word} = 10\ 1100\ 1100\ 1101$$

(b)

$$C_1 = \text{XOR}(1,3,5,7,9,11,13) = 1 \oplus 1 \oplus 0 \oplus 1 \oplus 0 \oplus 1 \oplus 0 = 0$$

$$C_2 = \text{XOR}(2,3,6,7,10,11,14) = 0 \oplus 1 \oplus 0 \oplus 1 \oplus 0 \oplus 1 \oplus 1 = 0$$

$$C_4 = \text{XOR}(4,5,6,7,12,13,14) = 1 \oplus 0 \oplus 0 \oplus 1 \oplus 1 \oplus 0 \oplus 1 = 0$$

$$C_8 = \text{XOR}(8,9,10,11,12,13,14) = 1 \oplus 0 \oplus 0 \oplus 1 \oplus 1 \oplus 0 \oplus 1 = 0$$

$$C = 0000. \text{ No Error.}$$

(c) Error in bit D_9 :

Received Code word : 10 1100 1110 1101

$$C_1 = \text{XOR}(1,3,5,7,9,11,13) = 1 \oplus 1 \oplus 0 \oplus 1 \oplus 1 \oplus 1 \oplus 0 = 1$$

$$C_2 = \text{XOR}(2,3,6,7,10,11,14) = 0 \oplus 1 \oplus 0 \oplus 1 \oplus 0 \oplus 1 \oplus 1 = 0$$

$$C_4 = \text{XOR}(4,5,6,7,12,13,14) = 1 \oplus 0 \oplus 0 \oplus 1 \oplus 1 \oplus 0 \oplus 1 = 0$$

$$C_8 = \text{XOR}(8,9,10,11,12,13,14) = 1 \oplus 1 \oplus 0 \oplus 1 \oplus 1 \oplus 0 \oplus 1 = 1$$

$$C = 1001 = 9. \text{ Hence error in bit 9 detected.}$$

(d) $P_{15} = \text{XOR}(1 \dots 14) = 0$, Hence, 15 bit word is 101 1001 1001 1010

Error in bits P_4 and D_{11} .

Received Code word : 101 0001 1000 1010

$$C_1 = \text{XOR}(1,3,5,7,9,11,13) = 1 \oplus 1 \oplus 0 \oplus 1 \oplus 0 \oplus 0 \oplus 0 = 1$$

$$C_2 = \text{XOR}(2,3,6,7,10,11,14) = 0 \oplus 1 \oplus 0 \oplus 1 \oplus 0 \oplus 0 \oplus 1 = 1$$

$$C_4 = \text{XOR}(4,5,6,7,12,13,14) = 0 \oplus 0 \oplus 0 \oplus 1 \oplus 1 \oplus 0 \oplus 1 = 1$$

$$C_8 = \text{XOR}(8,9,10,11,12,13,14) = 1 \oplus 0 \oplus 0 \oplus 0 \oplus 1 \oplus 0 \oplus 1 = 1$$

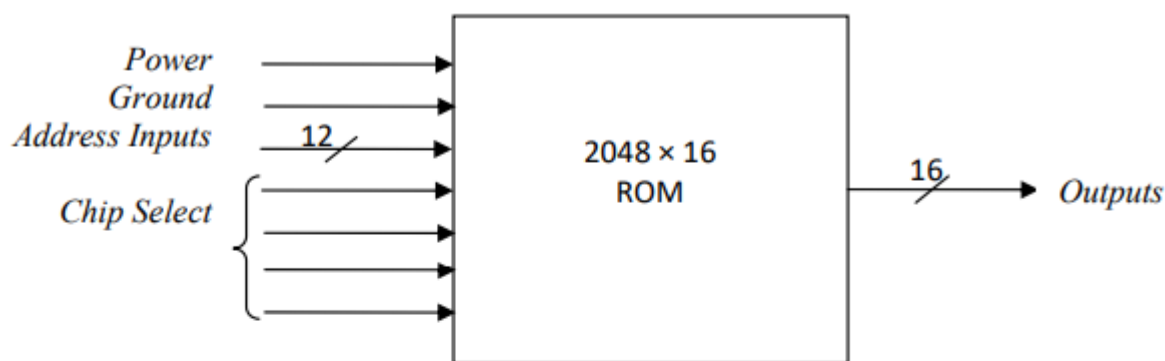
$$P_{15} = \text{XOR}(1 \dots 14) = 0$$

$$C \neq 0 \text{ and } P = 0. \text{ Indicates double error.}$$

7.15 Number of 128×8 ROM chips required to construct a 2048×8 ROM is $2048/128 = 16$
Additionally a 4×16 decoder is required to select among the 16 ROM chips.

7.16 Number of input address lines is 12, since $2^{12} = 4096$.

$$\begin{aligned} \text{Total number of pins needed} &= 12(\text{input address lines}) + 1 \text{ pin (power supply)} \\ &\quad + 1 (\text{ground}) + 4(\text{chip select}) + 16(\text{outputs}) = 34 \end{aligned}$$



7.17

Input Address	Output of ROM			
$I_5 I_4 I_3 I_2 I_1$	$D_6 D_5 D_4$	$D_3 D_2 D_1$	$D_0 (2^0)$	Decimal
0 0 0 0 0	0 0 0	0 0 0	0, 1	0, 1
0 0 0 0 1	0 0 0	0 0 1	0, 1	2, 3
...	...	...	...	...
...	...	...	...	... 16,
0 1 0 0 0	0 0 1	0 1 1	0, 1	17
0 1 0 0 1	0 0 1	1 0 0	0, 1	18, 19
...	...	...	...	...
...	...	...	...	... 60,
1 1 1 1 0	1 1 0	0 0 0	0, 1	61
1 1 1 1 1	1 1 0	0 0 1	0, 1	62, 63

7.18 (a) A converter that converts a 4-bit BCD number to Excess – 3 code

Inputs				Outputs			
I_3	I_2	I_1	I_0	O_3	O_2	O_1	O_0
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0

We see that 4 inputs and 4 outputs are needed. However, since O_0 is the complement of I_0 , we can simply connect an inverter to I_0 to generate O_0 . Thus we require to generate only 3 outputs through ROM, i.e. the minimum size of ROM should have 4 inputs and 3 outputs. Thus 16×3 ROM is required.

(b) A circuit that adds three 2-bit numbers along with an input carry-

The number of input bits = $2 \times 3 + 1 = 7$

The largest number that the output holds is $(3 + 3 + 3) + 1 = 10_{10} = 1010_2$. So the number of output bits is 4.

Hence ROM size is 128×4

(c) A circuit that multiplies three 4-bit numbers –

The number of input bits = $3 \times 4 = 12$

The largest number that the output holds is

$15 \times 15 \times 15 = 3375_{10} = 110100101111_2$. So the number of output bits is 12.

Hence ROM size is 4096×12 .

(d) A 4-bit binary incrementor / decrementor –

The number of input bits is 5 (4 for the number and 1 to determine increment or decrement operation). The number of output bits is 4.

Hence, ROM size is 32×4 .

7.19 $A(x, y, z) = \Sigma(0, 1, 5, 7)$

		y			
		00	01	11	10
x \ yz	0	m_0 1	m_1 1	m_3 0	m_2 0
	1	m_4 0	m_5 1	m_7 1	m_6 0

$$A = x'y' + xz$$

$$A' = x'y + xz'$$

$B(x, y, z) = \Sigma(2, 4, 5, 6)$

		y			
		00	01	11	10
x \ yz	0	m_0 0	m_1 0	m_3 0	m_2 1
	1	m_4 1	m_5 1	m_7 0	m_6 1

$$B = xy' + yz'$$

$$B' = x'y' + yz$$

$C(x, y, z) = \Sigma(0, 1, 2, 3, 4)$

		y			
		00	01	11	10
x \ yz	0	m_0 1	m_1 1	m_3 1	m_2 1
	1	m_4 1	m_5 0	m_7 0	m_6 0

$$C = x' + y'z'$$

$$C' = xy + xz$$

$$D(x, y, z) = \Sigma(3, 6, 7)$$

		y			
		00	01	11	10
yz \ x	0	m_0 0	m_1 0	m_3 1	m_2 0
	1	m_4 0	m_5 0	m_7 1	m_6 1
		z			

$$D = xy + yz$$

$$D' = y' + x'z'$$

If the outputs, A , B' , C' and D are considered, then we will get four product terms for implementation using PLA, $x'y'$, xz , yz , xy .

PLA program Table:

	Product term	Inputs			Outputs			
		x	y	z	A	B'	C'	D
$x'y'$	1	0	0	—	1	1	—	—
xz	2	1	—	1	1	—	1	—
yz	3	—	1	1	—	1	—	1
xy	4	1	1	—	—	—	1	1

Note : The outputs B' and C' are complemented to get the results B and C .

7.20

Inputs				Outputs			
w	x	y	z	A	B	C	D
0	0	0	0	1	0	1	1
0	0	0	1	0	0	0	1
0	0	1	0	1	0	0	1
0	0	1	1	0	1	0	0
0	1	0	0	0	0	1	1
0	1	0	1	1	1	0	0
0	1	1	0	0	0	0	0
0	1	1	1	1	1	0	1
1	0	0	0	1	0	1	0
1	0	0	1	0	1	0	1
1	0	1	0	0	0	0	0
1	0	1	1	0	1	0	0
1	1	0	0	0	0	1	0
1	1	0	1	0	1	0	0
1	1	1	0	1	0	0	0
1	1	1	1	0	1	0	0

$$M[0101]=1100$$

$$M[1111]=0100$$

7.21 Note: See truth table in Fig. 7.12(b).

$A_1 A_0$		A_1			
		00	01	11	10
A_2	0	m_0 0	m_1 0	m_3 0	m_2 0
	1	m_4 0	m_5 0	m_7 1	m_6 1

$$F_1 = A_2 A_1$$

$$F_1' = A_2' + A_1'$$

$A_1 A_0$		A_1			
		00	01	11	10
A_2	0	m_0 0	m_1 0	m_3 0	m_2 0
	1	m_4 1	m_5 1	m_7 1	m_6 0

$$F_2 = A_2 A_1' + A_2 A_0$$

$$F_2' = A_2' + A_1 A_0'$$

$A_1 A_0$		A_1			
		00	01	11	10
A_2	0	m_0 0	m_1 0	m_3 1	m_2 0
	1	m_4 0	m_5 1	m_7 0	m_6 0

$$F_3 = A_2' A_1 A_0 + A_2 A_1' A_0$$

$$F_3' = A_0' + A_2' A_1' + A_2 A_1$$

$A_1 A_0$		A_1			
		00	01	11	10
A_2	0	m_0 0	m_1 0	m_3 0	m_2 1
	1	m_4 0	m_5 0	m_7 0	m_6 1

$$F_4 = A_1 A_0'$$

$$F_4' = A_1' + A_0$$

Product term		Inputs		Outputs			
		$A_2 A_1 A_0$		F_1	F_2	F_3	F_4
$A_2 A_1$	1	1	1	-	1	-	-
A_2'	2	0	-	-	-	1	-
$A_1 A_0'$	3	-	1	0	-	1	1
$A_2' A_1 A_0$	4	-	1	1	-	-	1
$A_2 A_1'$	5	1	0	1	-	-	1
				<u>T C T T</u>			

Alternative: F_1', F_2', F_3, F_4
(5 terms)

7.22

Decimal	w	x	y	z	b_7	b_6	b_5	b_4	b_3	b_2	b_1	b_0
0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0	0	0	0	1
2	4	0	0	1	0	0	0	0	0	1	0	0
3	9	0	0	1	1	0	0	0	1	0	0	1
4	16	0	1	0	0	0	0	1	0	0	0	0
5	25	0	1	0	1	0	0	1	1	0	0	1
6	36	0	1	1	0	0	1	0	0	1	0	0
7	49	0	1	1	1	0	0	1	1	0	0	1
8	64	1	0	0	0	0	1	0	0	0	0	0
9	81	1	0	0	1	0	1	0	1	0	0	1
10	100	1	0	1	0	0	1	1	0	0	1	0
11	121	1	0	1	1	0	1	1	1	0	0	1
12	144	1	1	0	0	1	0	0	1	0	0	0
13	169	1	1	0	1	1	0	1	0	1	0	1
14	196	1	1	1	0	1	1	0	0	0	1	0
15	225	1	1	1	1	1	1	1	0	0	0	1

Note: $b_0 = z$, and $b_1 = 0$.
ROM would have 4 inputs
and 6 outputs. A 4 x 8
ROM would waste two
outputs.

		yz		y	
		00	01	11	10
wx	00	m_0	m_1	m_3	m_2 1
	01	m_4	m_5	m_7	m_6 1
w	11	m_{12}	m_{13}	m_{15}	m_{14} 1
	10	m_8	m_9	m_{11}	m_{10} 1

$$b_2 = yx'$$

		yz		y	
		00	01	11	10
wx	00	m_0	m_1	m_3	m_2
	01	m_4 1	m_5 1	m_7 1	m_6
w	11	m_{12} 1	m_{13}	m_{15}	m_{14}
	10	m_8	m_9 1	m_{11} 1	m_{10}

$$b_4 = w'xz + xy'z' + wx'z$$

		yz		y	
		00	01	11	10
wx	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7	m_6
w	11	m_{12}	m_{13}	m_{15} 1	m_{14} 1
	10	m_8 1	m_9 1	m_{11} 1	m_{10} 1

$$b_6 = wy + wx'$$

		yz		y	
		00	01	11	10
wx	00	m_0	m_1	m_3 1	m_2
	01	m_4	m_5 1	m_7	m_6
w	11	m_{12}	m_{13} 1	m_{15}	m_{14}
	10	m_8	m_9	m_{11} 1	m_{10}

$$b_3 = xy'z + x'yz$$

		yz		y	
		00	01	11	10
wx	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7 1	m_6 1
w	11	m_{12}	m_{13} 1	m_{15} 1	m_{14}
	10	m_8	m_9	m_{11} 1	m_{10} 1

$$b_5 = w'xy + wxz + wx'y$$

		yz		y	
		00	01	11	10
wx	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7	m_6
w	11	m_{12} 1	m_{13} 1	m_{15} 1	m_{14} 1
	10	m_8	m_9	m_{11}	m_{10}

$$b_7 = wx$$

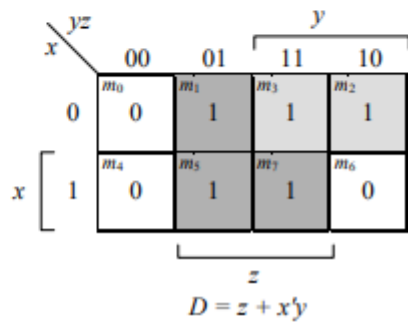
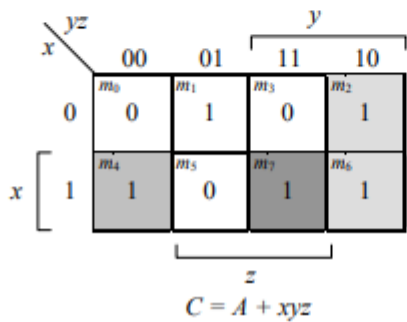
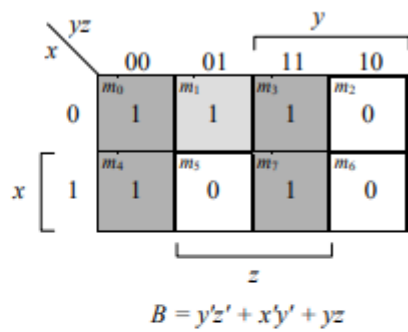
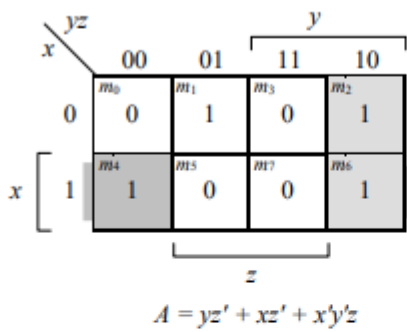
7.23

From Fig. 4-3:
 $w = A + BC + BD$
 $w' = A'B' + A'C'D'$
 $x = B'C + B'D + BC'D'$
 $x' = B'C'D' + BC \ BD$
 $y = CD + C'D'$
 $y' = C'D + CD'$
 $z = D'$
 $z' = D$
Use w, x', y, z (/ terms)

Product term		Inputs				Outputs			
		A	B	C	D	F ₁	F ₂	F ₃	F ₄
A	1	1	-	-	-	1	-	-	-
BC	2	-	1	1	-	1	1	-	-
BD	3	-	1	-	1	1	1	-	-
B'C'D'	4	-	0	0	0	-	1	-	-
CD	5	-	-	1	1	-	-	1	-
C'D'	6	-	-	0	0	-	-	1	-
D'	7	-	-	-	0	-	-	-	1
						T	C	T	T

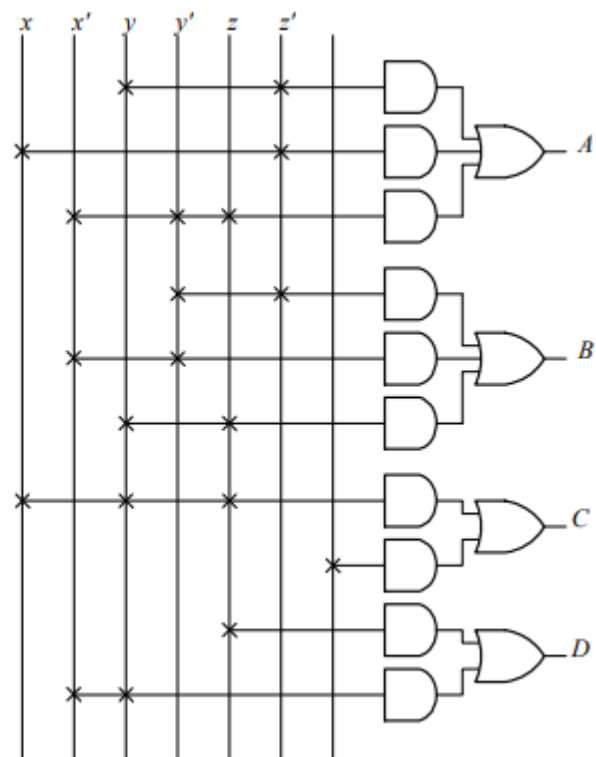
7.24

Product term	AND	Inputs				Outputs
		A	B	C	D	
1		1	-	-	-	$w = A + BC + BD$
2		-	1	1	-	
3		-	1	-	1	
4		-	0	1	-	$x = B'C + B'D + BC'D'$
5		-	0	-	1	
6		-	1	0	0	
7		-	-	1	1	$y = CD + C'D'$
8		-	-	0	0	
9		-	-	-	-	
10		-	-	-	0	$z = D'$
11		-	-	-	-	
12		-	-	-	-	

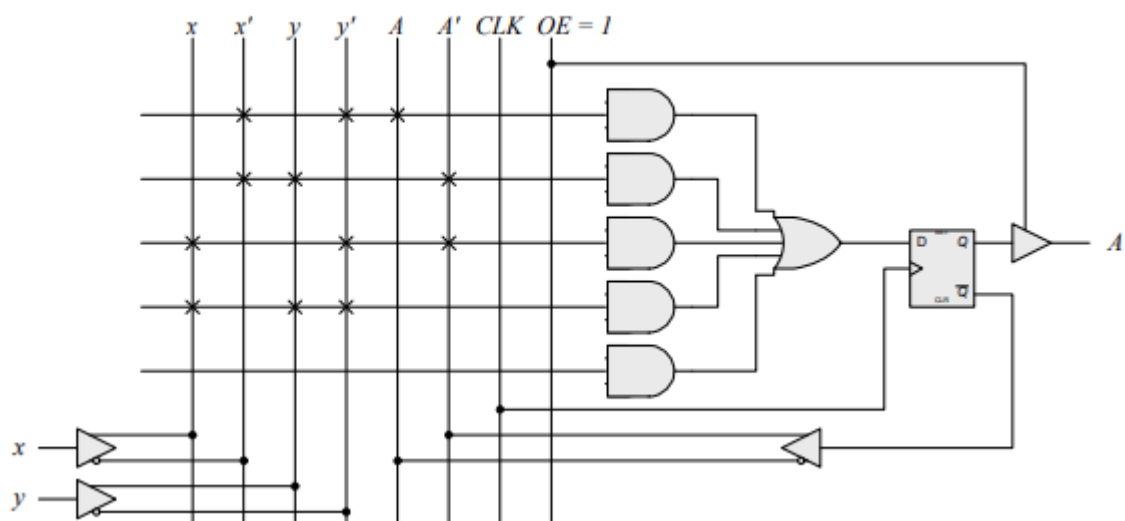


AND				
Product	Inputs			Outputs
term	x	y	z	
1	-	1	0	$A = yz' + xz' + x'y'z$
2	1	-	0	
3	0	0	1	
4	-	0	0	$B = y'z' + x'y' + yz$
5	0	0	-	
6	0	1	1	
7	0	-	1	$C = A + xyz$
8	1	1	1	
9	0	-	-	
10	0	-	1	$D = z + x'y$
11	0	1	-	
12	-	-	-	

$A = yz' + xz' + x'y'z$
 $B = y'z' + x'y' + yz$
 $C = A + xyz$
 $D = z + x'y$



7.26



7.27

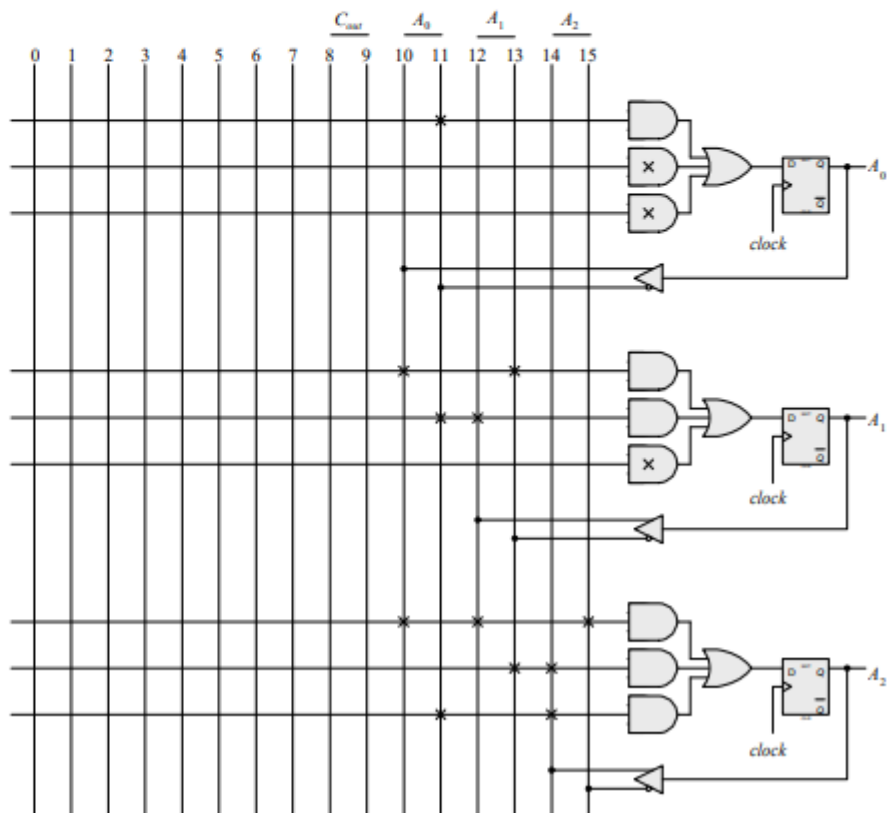
The results of Prob. 6.17 can be used to develop the equations for a three-bit binary counter with D-type flip-flops.

$$DA_0 = A'_0$$

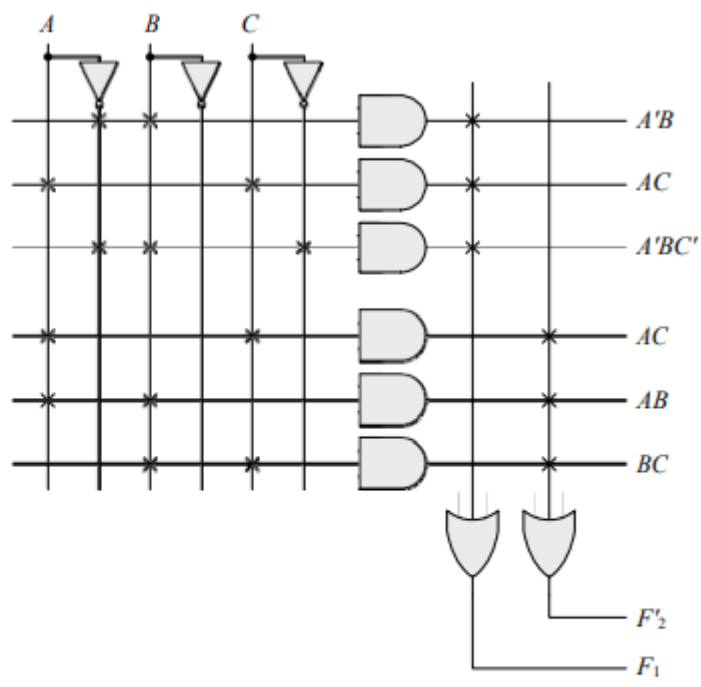
$$DA_1 = A'_1A_0 + A_1A'_0$$

$$DA_2 = A'_2A_1A_0 + A_2A'_1 + A_2A'_0$$

$$C_{out} = A_2A_1A_0$$



7.28



7.29

	Product	Inputs			Output
	term	x	y	A	D_A
$x'y'A$	1	0	0	1	1
$x'yA'$	2	0	1	0	1
$xy'A'$	3	1	0	0	1
xyA	4	1	1	1	1